debugging proof

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assessment / task: parking lot tracker - part a
date: 2026-09-12

bug 1 - empty lot totals wrong

1. what was the problem?

expected behaviour: total bays 3, occupied bays 0, free bays 3
actual behaviour: total bays 3, occupied bays 1, free bays 2

2. evidence of the bug

```
star@legacystarpad:~/Documents/Github/ezparking$ python3 parking_lot_tracker_part_a.py
choose an option: 4
total bays: 3
occupied bays: 1
free bays: 2
```

3. what caused the problem?

`relevant code: i_occupied = 1`

explanation: the counter started at one, so the empty lot was counted as having one occupied bay. it should start at zero.

4. how did you debug it?

i tested calculateTotals() with three free bays and compared expected and actual output.

5. what did you change?

`corrected code: i_occupied = 0`

the counter now starts at zero before counting occupied bays.

6. proof that the fix works

test performed: empty car park totals expected: total 3, occupied 0, free 3 result: pass

```
star@legacystarpad:~/Documents/Github/ezparking$ python3 parking_lot_tracker_part_a.py
choose an option: 4
total bays: 3
occupied bays: 0
free bays: 3
```

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bug 2 - removed bay stays occupied

1. what was the problem?

expected behaviour: plate cleared and bay 1 marked free
actual behaviour: plate cleared but bay 1 remained marked occupied

2. evidence of the bug

```
star@legacystarpad:~/Documents/Github/ezparking$ python3 parking_lot_tracker_part_a.py
choose an option: 2
enter bay number: 1
car removed
choose an option: 3
bay 1: occupied -
```

3. what caused the problem?

`relevant code: dictBays[bay_number]["occupied"] = True`
explanation: the removal code cleared the plate but assigned true to occupied. the bay should be marked false.

4. how did you debug it?

i tested entry, removal, and status display, then compared the expected free state with the actual state.

5. what did you change?

`corrected code: dictBays[bay_number]["occupied"] = False`
a removed car now leaves its bay free.

6. proof that the fix works

test performed: remove car from bay 1 and view status expected: bay 1 free result: pass

```
star@legacystarpad:~/Documents/Github/ezparking$ python3 parking_lot_tracker_part_a.py
choose an option: 2
enter bay number: 1
car removed
choose an option: 3
bay 1: free
```

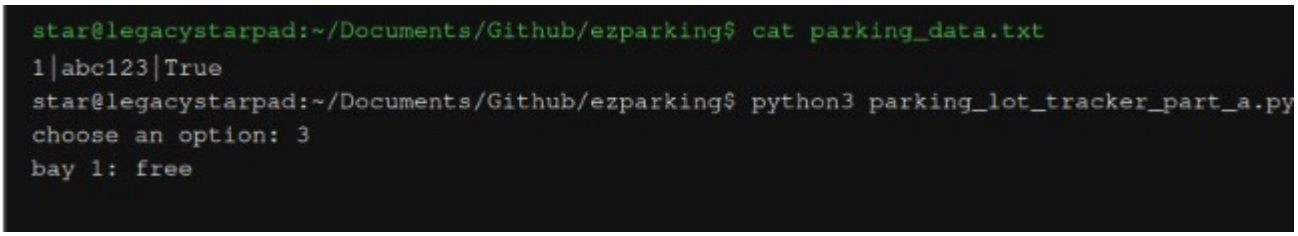
debugging proof

bug 3 - saved occupied bay reloads as free

1. what was the problem?

expected behaviour: bay 1 stays occupied with plate abc123 after save and reload
actual behaviour: bay 1 reloaded as free

2. evidence of the bug



3. what caused the problem?

`relevant code: occupied_text == "true"`
explanation: loadData() compared occupied_text with lowercase true, but saveData() wrote python True with capital t.

4. how did you debug it?

i compared the value written to parking_data.txt with the value checked by loadData().

5. what did you change?

`corrected code: dictBays[bay_number]["occupied"] = occupied_text == "True"`
the load check now matches the saved boolean string.

6. proof that the fix works

test performed: saved occupied bay, closed and reloaded expected: bay 1 occupied result: pass



final debugging reflection

which bug was the most difficult to solve, and why?
the reload bug was most difficult because saving and loading used different boolean text values.

what debugging technique was most useful?
comparing saved data with the loading condition was most useful.

what did you learn that could help you avoid or identify similar bugs in future?
i learned to test empty, occupied, removed, and reloaded states and to check both sides of file handling.